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Paleomagnetism of an east-west transect across the Cascade arc in southern Washington: Implications for regional tectonism

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Abstract. Paleomagnetic data from a transect across the Cascade arc in southern Washington were collected to test models of vertical axis rotations for this part of the Pacific Northwest and to provide information on the tectonic history of the St. Helens seismic zone. The 75 site-mean directions are divided into three groups based on isotopic, fission track, and stratigraphic ages. Group 1 consists of samples from rocks deposited between 39 and 30 Ma and shows a mean clockwise vertical axis rotation of 34° ± 13°, group 2 consists of samples from rocks deposited between 30 and 24 Ma and shows a mean clockwise rotation of 17° ± 11°, and group 3 consists of samples from rocks between 24 and 16 Ma and shows a mean clockwise rotation of 20° ± 12°. Although these three values of rotation are statistically indistinguishable at the 95% confidence level, we interpret them in combination with other data to indicate differential rotation across the St. Helens seismic zone (SHZ). The available paleomagnetic data for Eocene-Oligocene rocks west of the seismic zone show clockwise vertical axis rotations of 30° ± 8° and 35° ± 9° consistent with the value for group 1. The rotational values for groups 2 and 3 and the intrusive suite of Kidd Creek (13 Ma), despite their different ages, have similar values east of the SHZ. Comparing these groups of data east and west of the seismic zone indicates a differential rotation of 10° ± 3° across it. In addition, the rates of rotation in southern Washington are similar to those for rocks in the western Cascades of Oregon and indicate that rotational deformation of Miocene age extends northward into southern Washington and eastward into the axis of the arc.

1. Introduction

Paleomagnetic studies of Cenozoic rocks in the Pacific Northwest have been instrumental in documenting clockwise vertical axis rotations of crustal blocks within the convergent continental margin [*Simpson and Cox*, 1977; *Beck*, 1980; *Wells et al.*, 1984]. Distributed results from the forearc, arc, and back arc regions have indicated that rotations tend to increase with age and proximity to the coast [*Wells and Heller*, 1988; *Wells et al.*, 1989], apparently resulting from dextral shear caused by oblique subduction of the Juan de Fuca plate [*Beck*, 1980; *England and Wells*, 1991]. Rotations have been accommodated within the upper crust of the arc and forearc regions by motion along intermediate-scale and smaller strike-slip faults [*Wells and Coe*, 1985]. The increased magnitude of coastal rotations southward from Washington into Oregon, however, suggests an equivalent contribution from extensional deformation in regions outboard of the Basin and Range Province [*Magill and Cox*, 1981; *Grommé et al.*, 1986; *Wells*, 1990]. Eastward migration of the volcanic front away from older magmatic belts in western Oregon as well as westward migration of bimodal Basin and Range volcanism behind the arc are consistent with the paleomagnetically-determined rotations of the arc terrane [*Wells*, 1990].

In general, Eocene and Oligocene rocks of the southern Washington and Oregon Coast Ranges have been rotated between ~20° and 70° clockwise, coeval rocks in the Western Cascades have been rotated between 18° and 36° clockwise [*Grommé et al.*, 1986], and middle Miocene Columbia River Basalt in the Western Cascades (15-12 Ma) has been rotated between 10° and 15° clockwise [*Wells et al.*, 1989]. In addition, paleomagnetic data bearing on the question of vertical axis rotations in the southern Washington Cascades are available from several investigations. *Beck and Burr* [1979] determined a mean clockwise rotation of 30° ± 8° for a large area underlain by late Eocene to Oligocene volcanic rocks west and southwest of Mount St. Helens, which they termed the "Goble block." Working in coeval rocks immediately to the east, between Mount St. Helens and the Columbia River, *Bates et al.* [1981] obtained a similar mean rotation of 35° ± 9° for strata they correlated with the Ohanapecosh Formation. More recently, *Hagstrum et al.* [1998] have determined a 22° ± 8°

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clockwise rotation for the intrusive suite of Kidd Creek which has K-Ar and fission track ages of ~13 Ma. This rotation of the Kidd Creek intrusive suite is larger than expected for rocks of this age in the region and suggests that much of the rotational deformation in southern Washington may be younger than previously thought.

Recently, extensive mapping and new age determinations across the Cascade arc in southern Washington have begun to clarify the complex stratigraphic and structural relationships within this terrane [see *Evarts and Swanson*, 1994]. Previously one of the most poorly mapped and least understood areas in the Pacific Northwest, the volcanogenic rocks are characterized by rapid facies changes, widespread distribution of similar lithologies, and few marker beds. This paleomagnetic investigation was undertaken to collect samples from stratigraphically known localities across the arc, particularly from dated units, to determine the tectonic evolution of this region and its role in the deformation of the margin as a whole. In particular, our intent was to ascertain whether older rocks in the vicinity of the intrusive suite of Kidd Creek have been rotated an equivalent or greater amount than the younger rocks and what contribution the poorly understood St. Helens seismic zone has made to the structural development of the southern Washington Cascades.

2. Geologic Setting

The Cascade volcanic arc has resulted from subduction along the Pacific Northwest coast, from northern California to southern British Columbia, over the last 40 m.y. Presently, there is considerable variation in the along-strike tectonic setting, the type of volcanism, and the past structural and petrologic evolution of the arc [*Smith*, 1993]. In northern Washington a few scattered Quaternary statovolcanoes rest on pre-Tertiary crust composed of allochthonous terranes accreted to North America in the late Mesozoic. In Oregon, composite volcanoes rest on a broad platform of Pliocene and Holocene mafic lava flows which occupy, in part, a north-south trending graben. The southern Washington Cascades (Figure 1) are a transitional zone in which Quaternary composite volcanoes are unconformably above folded and altered volcanic rocks of late Eocene to middle Miocene age. Between the volcanoes Tertiary strata are partially covered by Pliocene and younger, mostly mafic, volcanic rocks that occur in small isolated volcanic fields and as scattered monogenetic vents [*Evarts and Swanson*, 1994; *Guffanti and Weaver*, 1988; *Smith*, 1993].

The crustal structure beneath the Cascade arc in southern Washington is poorly understood, and geophysical evidence indicates it is much more complex than that beneath the Oregon Cascades. The difference is most likely due to multiple accretionary structures in southern Washington formed during late Mesozoic and early Tertiary plate convergence [*Stanley et al.*, 1990]. Regional magnetotelluric surveys indicate a high-conductivity crustal anomaly between Mount St. Helens and Mount Rainier which has been named the southern Washington Cascades conductor (SWCC, Figure 2). This anomalous region is up to 15 km thick, and *Stanley et al.* [1992] interpret it as a sequence of Paleogene marine sedimentary rocks deposited within a pre-Cascade forearc basin, accretionary prism, or pull-apart basin.

Gravity surveys show a broad isostatic residual gravity trough coincident with the Cascade Range that is interrupted in southern Washington by a northwest trending gravity high (Figure 2). *Blakely and Jachens* [1990] suggest that this feature reflects the juxtaposition of large fault-bounded crustal blocks of contrasting densities. To the west of the Cascade arc gravity low is a belt of intense gravity and aeromagnetic highs associated with the early Tertiary oceanic basaltic basement of the Coast Range [*Blakely and Jachens*, 1990; *Finn*, 1990]. *Finn* [1990] has interpreted these data as indicating extension of the Coast Range basaltic basement inland beneath the volcanic arc at shallow depths as far east as Mount St. Helens.

Most earthquakes in the southern Washington Cascades occur along the northwest trending, 90-km-long St. Helens seismic zone (SHZ, Figure 2). Their foci are generally 5 to 15 km deep, and focal mechanisms indicate dextral slip on near-vertical fault planes parallel to the seismic zone; these slip vectors are consistent with the regional compression oriented N23°E [*Weaver and Smith*, 1983; *Ma et al.*, 1992]. The SHZ roughly coincides with the western limit of the SWCC and the inferred eastern limit of oceanic basement and may mark the boundary between these crustal blocks. There is, however, no evidence of such a major structure in the overlying arc rocks, although the area has not been mapped well enough to be certain that the zone has no surface expression. Another northwest trending seismic zone without clear surface expression is the Western Rainier seismic zone, which is better defined at its northern end west of Mount Rainier (Figure 2).

The structure of the southern Washington Cascades is dominated by broad, open folds trending north to northwest [*Hammond*, 1980; *Walsh et al.*, 1987]. The folds tend to be more closely spaced and have steeper limbs in the northeast part of the area, and their axes often extend for more than 20 km [*Evarts and Swanson*, 1994]. Folding appears to have occurred after 19 Ma and before 12 Ma as indicated by a paleomagnetic "tilt" test and fission track and ^{40}Ar-^{39}Ar dates for the intrusive suite of Kidd Creek and its host rocks [*Hagstrum et al.*, 1998]. Faults are less prominent than folds in the region and can rarely be traced for more than 10 km. Most faults dip steeply and strike northwest or northeast and are much more abundant west of Mount St. Helens [*Evarts and Swanson*, 1994; R. C. Evarts, unpublished mapping, 1991-95]. The direction of offset and amount of displacement on these faults usually cannot be determined, but they are likely strike-slip faults that accommodated the paleomagnetically documented vertical axis rotations in the western Cascades [*Wells and Coe*, 1985].

Stratigraphic nomenclature of the Tertiary arc rocks is unsettled. *Fiske et al.* [1963] formally defined three middle Tertiary units in Mount Rainier National Park: the Ohanapecosh, Stevens Ridge, and Fifes Peak Formations. Elsewhere in the southern Washington Cascades, rocks that are broadly correlative with these formations have been assigned to the Goble Volcanics and the Hatchet Mountain Formation [*Wilkinson et al.*, 1946; *Roberts*, 1958; *Phillips*, 1987]. Recent detailed mapping of intervening areas (cited by *Evarts and Swanson* [1994]) shows that stratigraphic relations within the arc are more complex and that the formal stratigraphic units generally cannot be extended much beyond their type areas. Additional mapping is needed before a formal, regionally consistent, stratigraphy can be adopted.

All rocks of the Tertiary Cascades in southern Washington have been affected by zeolite-facies metamorphism resulting from burial in the high-heat flow environment of the arc.

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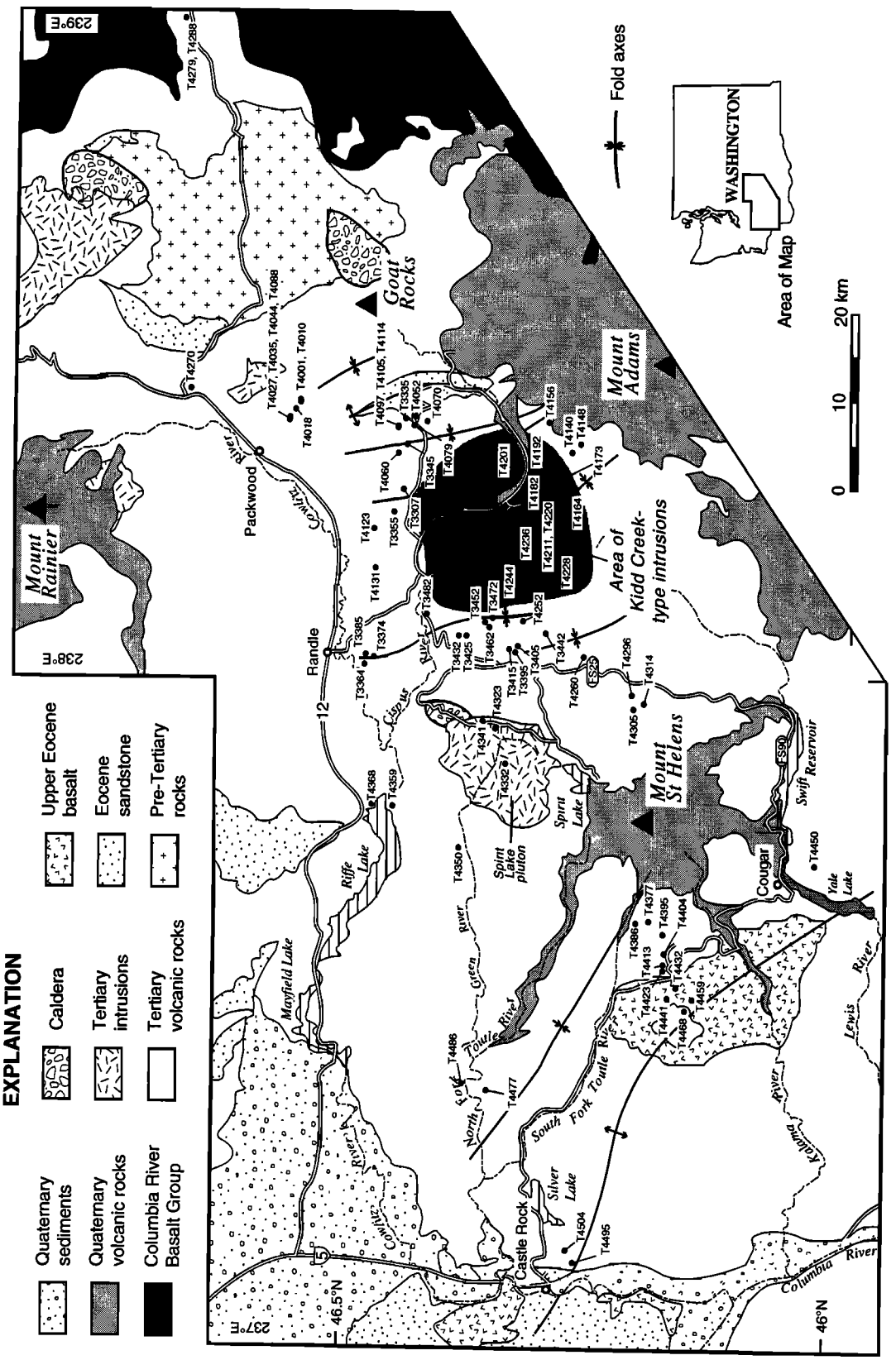


Figure 1. Simplified geologic map of the western Cascade Range in southern Washington (inset), after *Evarts and Swanson* [1994], showing major streams, selected geologic units, fold axes, and paleomagnetic sampling sites.

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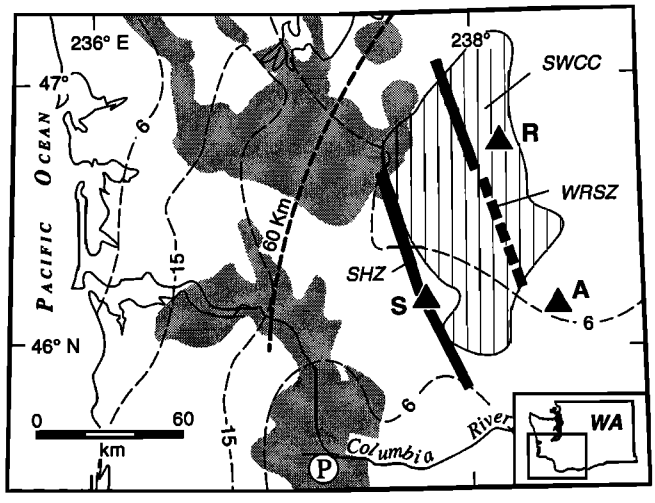


Figure 2. Map of southwestern Washington (inset) showing geophysical and tectonic elements after *Evarts and Swanson* [1994], including the St. Helens seismic zone (SHZ), Western Rainier seismic zone (WRSZ), outline of the southern Washington Cascades conductance anomaly (SWCC) at the 2000-S contour, positive magnetic anomalies (shaded areas), 6- and 15-mGal isostatic residual gravity contours, and 60-km depth contour on top of the Juan de Fuca plate. A, Mount Adams; circled P, Portland; R, Mount Rainier; and S, Mount St. Helens.

Overall, the intensity of this metamorphism is greatest in the center of the arc where intrusions are most abundant. Local contact metamorphism and hydrothermal alteration associated with subvolcanic intrusions are also widespread in the region [*Evarts and Swanson*, 1994].

3. Methods and Results

Drill core samples of mainly volcanic and intrusive rocks were collected at 73 sites and were oriented with both magnetic and solar compasses. Usually, between 8 and 10 samples were collected per site. Areas of extensive alteration were avoided, and samples from only the freshest exposures were collected. Structural attitudes for host rocks are based on flow contacts, interbedded sediments where available, or regional values for extrusive rocks. The remanent magnetization of individual specimens, cut from the core samples, was measured with a cryogenic magnetometer. Specimens from all samples were progressively demagnetized with standard alternating field (AF) and thermal procedures to peak inductions of 100 mT and maximum temperatures of 675°C, respectively. Lines representing the characteristic and overprint magnetization directions, or planes for two overlapping components, were fitted to the demagnetization data using a least squares method based on principal component analysis [*Kirschvink*, 1980]. The statistics of *Fisher* [1953] were used in analyzing the line-fit site and overall mean directions, and Bingham statistics [*Onstott*, 1980] were used in analyzing the intersecting planes (Table 1).

Representative orthogonal vector plots of demagnetization data for rocks of the southern Washington Cascades are shown in Figure 3. In general, a majority of sites have well-grouped natural remanent magnetization (NRM) directions and show univectorial behavior upon demagnetization (Figures 3a and 3c); NRM intensities usually range between 1 and 10^{-2} A m^{-1}. A consistent normal-polarity overprint component was often removed during AF and thermal demagnetization below 30 mT and 250°C (Figure 3b) and is most apparent in specimens with reversed-polarity characteristic magnetizations (Figures 3d, 3e, and 3f). At some sites, specimens contain both normal- and reversed-polarity characteristic magnetizations (Table 1). At several other sites the data are also indicative of a second component of magnetization at the high end of the demagnetization curve, but the signal is too noisy to be fitted with either least squares lines or planes.

The thermal and AF demagnetization data are consistent with magnetite being the remanence-carrying magnetic mineral in most of these rocks. In samples from ~10 to 20% of the sites, however, the demagnetization data indicate that hematite with blocking temperatures >600°C, apparently in varying amounts, is also present in these samples. In most cases, there is no difference in the characteristic direction of magnetization between the two mineral phases within a given specimen (Figure 3e), but in some cases, magnetite appears to carry a component of one polarity and hematite appears to carry a second, higher-temperature component of the opposite polarity.

At two sites (T4001 and T4010) an andesite breccia was sampled to determine if these rocks had been formed at temperatures high enough (above the blocking temperatures of their constituent magnetic minerals) to have a consistent direction of magnetization [see *Hoblitt and Kellogg*, 1979]. The specimens from both sites have random magnetizations [*Watson*, 1956], indicating that the flows were brecciated at temperatures below their magnetic blocking temperatures. Highly scattered NRM directions and variable NRM intensities were found at several sites that probably have been affected by lightning strikes. Least squares planes were used to successfully determine the characteristic directions of magnetization in samples from these sites.

4. Discussion and Conclusions

4.1. Data Reliability

Directions of the normal-polarity overprint magnetization removed from the samples are shown in Figure 4. This magnetization does not coincide with the present-day field (PDF) in southern Washington (I = 68.8°, D = 19.6° [*Peddie*, 1987]) and therefore was not acquired under current geomagnetic field conditions. Apparently, it was acquired as a thermoviscous magnetization over a long enough period of the Brunhes polarity chron to average out paleosecular variation (PSV) of the geomagnetic field. The average direction for this component corresponds to the recent dipole field for southern Washington (I = 64.5°, D = 0°) and may result in part from burial, zeolite-facies metamorphism, and/or uplift of the Cascades [*Tolan and Beeson*, 1984; *Swanson*, 1997].

Sites from across the transect in southern Washington (Figure 1) have been divided into three groups. Group 1 includes sites in the oldest rocks that have been dated between about 39 and 30 Ma (Table 1); also included are sites stratigraphically associated with the dated rock units. Group 2 includes sites in units that were deposited between ~30 and 24 Ma, and group 3 includes sites in rocks deposited between ~24 and 16 Ma. Site-mean directions for the three groups with their 95% confidence limits are shown in Figure 5. Each group has

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Table 1. Paleomagnetic Data for Volcanic and Intrusive Rocks of the Cascade Arc, Southern Washington

Site	*Unit*	λ_S	ϕ_S	I_t	D_t	*Strike/Dip*	I_c	D_c	N/N_0	R	k	α_{95}	λ_P	ϕ_P
						Group 1 (39-30 Ma)								
T3307[a]	BA	46.445	238.260	-31.7	236.8	0/0	-31.7	236.8	10/10	9.8778	74	5.7	35.1	340.7
T3335	A	46.440	238.363	-57.2	149.5	280/08	-50.9	156.1	10/10	9.9110	101	4.8	66.4	117.7
T3345	A	46.440	238.329	66.0	12.6	290/07	59.0	14.1	08/10	7.9791	334	3.0	77.8	356.2
T3355	A	46.454	238.226	-50.9	219.3	280/07	-44.7	215.7	07/09	6.9165	72	7.2	55.4	351.2
T4350[b]	BA	46.395	237.764	-60.5	142.8	355/19	-65.3	179.8	09/09	8.3753	13	15.0	89.0	230.0
T4377[c]	BA	46.190	237.641	71.4	349.2	25/08	74.7	13.5	09/09	8.7669	34	8.9	73.0	260.1
T4386	B	46.208	237.644	-68.3	183.5	25/08	-69.9	204.2	09/09	8.9947	1512	1.3	72.8	292.6
T4395	A	46.174	237.616	73.1	2.4	325/15	61.5	26.1	09/09	8.9698	265	3.2	71.1	328.9
T4404[d]	A	46.174	237.589	62.4	235.1	320/25	86.6	273.5	09/09	8.9115	90	5.4	46.2	227.7
T4413	BA	46.173	237.576	77.4	29.1	330/25	53.6	49.1	10/10	9.9502	181	3.6	51.3	327.8
T4423	BA	46.171	237.569	75.6	344.9	315/20	60.2	19.3	09/09	8.9061	85	5.6	75.2	340.6
T4432	B	46.158	237.552	79.2	166.6	320/18	73.8	86.8	09/09	8.9395	132	4.5	40.0	278.5
T4441	B	46.165	237.539	59.7	32.4	320/18	42.3	38.1	09/09	8.9567	185	3.8	52.6	349.8
T4450	AT	46.023	237.732	-29.2	146.5	30/30	-54.4	162.0	07/09	6.7862	28	11.6	72.5	115.0
T4459	B	46.142	237.538	74.1	346.9	315/20	58.7	18.4	09/09	8.9633	218	3.5	75.0	347.5
T4468	BA	46.149	237.515	80.9	45.5	315/25	55.9	45.1	09/09	8.6132	21	11.6	55.3	327.3
T4468				-42.3	175.1	315/25	-24.1	186.7	05/09	4.9082	44	11.7	56.0	45.8
T4477	A	46.357	237.387	52.9	33.0	0/0	52.9	33.0	08/09	7.9797	345	3.0	61.9	342.9
T4486	A	46.382	237.393	-72.4	223.6	290/04	-68.7	219.4	09/09	8.9901	805	1.8	64.0	300.3
T4495	B	46.262	237.129	67.8	329.3	360/10	70.9	354.8	09/09	8.9286	112	4.9	80.4	219.4
T4504	BA	46.268	237.147	66.8	25.3	45/06	68.1	39.5	09/09	8.9238	105	5.1	63.8	301.7
Average				67.4	7.8	(k_G=14)	59.6	24.6	20/21	18.9811	k_S=19	7.8	72.2	331.1
						Group 2 (30-24 Ma)								
T4018	AT	46.558	238.377	70.4	30.8	290/25	45.6	25.2	09/09	8.8575	56	6.9	62.2	4.1
T4027	A	46.561	238.363	72.3	106.4	280/30	57.3	44.1	08/08	7.8989	69	6.7	56.7	327.1
T4035	A	46.562	238.361	74.0	88.1	280/30	53.5	37.0	09/09	8.8021	40	8.2	59.5	339.2
T4044	A	46.562	238.360	69.8	135.4	280/30	65.9	53.6	08/08	7.7904	33	9.7	54.4	305.6
T4052	A	46.433	238.363	76.8	8.5	305/11	66.5	20.2	07/08	6.8190	33	10.6	76.2	310.4
T4060	A	46.445	238.317	65.3	7.8	280/06	59.3	8.2	10/10	9.8926	84	5.3	81.3	12.5
T4060[d]				-66.2	338.1	280/06	-71.1	328.9	05/10	4.9887	353	4.1	15.3	255.9
T4070	A	46.424	238.356	-49.6	189.6	290/12	-37.8	191.5	08/09	7.9209	88	5.9	63.1	34.1
T4079	A	46.435	238.354	-78.7	212.6	290/08	-70.8	207.5	08/09	[-211, -2]		[3.0, 4.2]	70.7	291.5
T4088	A	46.561	238.359	67.3	293.3	300/35	51.4	352.0	09/09	8.8669	60	6.7	74.3	84.1
T4097	A	46.450	238.355	67.9	359.0	300/15	54.3	10.6	08/08	7.9031	72	6.6	75.9	20.0
T4105	A	46.450	238.354	33.6	330.0	300/15	25.3	337.1	09/09	[-35, -1]		[7.0, 10.1]	51.6	96.0
T4114	A	46.451	238.352	58.9	328.1	300/15	49.9	345.0	07/09	6.7754	27	11.9	70.4	99.9
T4182	A	46.283	238.278	57.1	35.2	130/12	69.0	32.6	04/10	3.9805	154	7.4	68.1	299.8
T4192[e]	A	46.286	238.280	41.8	2.5	130/12	50.8	354.1	09/09	8.5812	19	12.1	74.6	77.7
T4201	R	46.339	238.273	-41.1	224.4	130/30	-70.9	230.2	06/10	5.9900	501	3.0	57.8	293.5
T4211[f]	A	46.308	238.138	52.6	18.3	170/12	56.8	2.3	08/09	7.9731	261	3.4	80.9	46.7
T4220	A	46.308	238.137	61.3	18.4	170/12	64.9	355.5	08/08	7.9116	79	6.3	86.9	159.0
T4228	A	46.275	238.171	67.9	349.5	340/15	61.5	18.9	07/08	6.9438	107	5.9	76.1	336.7
T4236	A	46.315	238.150	74.8	19.9	170/12	76.2	332.2	08/08	7.9041	73	6.5	66.7	206.9
T4244	A	46.317	238.099	-62.0	170.3	0/0	-62.0	170.3	08/08	7.8581	49	8.0	82.5	27.3
T4252	A	46.318	238.068	-66.6	206.4	160/08	-71.6	189.8	08/08	7.6067	18	13.5	78.3	265.7
T4260	D	46.256	238.015	69.9	13.6	340/18	56.5	38.7	10/10	9.9661	266	3.0	60.0	331.8
T4270[g]	AT	46.668	238.407	60.4	50.6	180/30	70.7	341.6	08/09	7.9315	102	5.5	75.8	190.9
T4359	BA	46.469	237.803	-39.8	151.1	355/19	-45.0	168.0	07/09	6.9367	95	6.2	67.9	87.5
T4368[h]	BA	46.486	237.819	-23.2	148.1	355/19	-30.5	157.1	09/09	8.8751	64	6.5	54.4	97.7
Average				65.8	7.1	(k_G=12)	59.2	7.2	25/26	23.8247	k_S=20	6.6	81.9	0.3
						Group 3 (24-16 Ma)								
T3364[i]	B	46.484	238.005	-67.0	170.4	330/20	-54.9	200.4	10/10	9.9679	280	2.9	71.1	356.5
T3374[i]	B	46.476	238.011	-66.3	174.7	330/23	-51.1	204.5	11/11	10.9718	355	2.4	66.1	357.4
T3385[i]	B	46.484	238.014	58.2	16.7	330/20	41.9	31.0	08/10	7.8328	42	8.7	56.7	359.3
T3395[j]	DT	46.326	238.023	-47.7	180.6	355/15	-44.3	195.8	10/10	9.9860	643	1.9	66.1	20.9
T3405	A	46.325	238.029	-51.6	184.7	355/15	-46.9	201.4	10/10	9.8061	46	7.2	65.2	7.8
T3415	BA	46.331	238.028	49.2	335.8	355/12	51.8	350.1	10/10	9.9708	309	2.8	74.1	90.2
T3425[d]	R	46.381	238.044	-31.7	120.1	350/07	-36.9	123.3	07/07	6.9890	545	2.6	37.5	138.7
T3432	A	46.385	238.045	71.4	283.6	345/05	75.6	292.9	10/10	9.9497	179	3.6	50.1	197.2
T3442[d]	A	46.297	238.047	1.5	19.7	345/05	-1.3	19.7	08/10	7.9422	121	5.1	40.0	32.0
T3452	BA	46.359	238.065	-60.9	204.5	340/03	-58.7	208.0	08/10	7.9711	242	3.6	68.5	336.7
T3462	BA	46.353	238.058	-55.1	159.0	340/05	-54.8	166.2	09/10	8.9460	148	4.2	74.9	286.4
T3472[d]	A	46.349	238.069	-79.3	273.5	340/03	-76.5	268.5	10/10	9.8986	89	5.2	41.3	273.2
T3482[k]	RD	46.421	238.076	51.1	30.4	0/0	51.1	30.4	10/10	9.8659	67	5.9	62.5	349.4
T4123	R	46.475	238.199	51.6	30.0	150/16	64.4	14.2	08/08	7.8206	39	9.0	80.2	324.9
T4131[d,l]	A	46.473	238.143	71.8	19.2	150/25	73.9	287.4	09/09	8.9588	194	3.7	47.0	193.9

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Table 1. (continued)

Site	Unit	λ_S	ϕ_S	I_t	D_t	Strike/Dip	I_c	D_c	N/N_0	R	k	α_{95}	λ_P	ϕ_P
Group 3 (24-16 Ma) **(continued)**														
T4140[d]	A	46.268	238.315	-24.5	158.3	0/0	-24.5	158.3	07/08	6.8511	40	9.6	51.8	94.1
T4148	A	46.261	238.327	82.5	318.7	100/02	83.6	304.8	08/08	7.9557	158	4.4	52.3	221.1
T4155[d,m]	AT	46.293	238.359	20.3	201.8	120/27	-6.5	202.3	06/08	5.9027	51	9.4	42.8	27.3
T4164	A	46.273	238.293	56.7	47.4	120/02	58.6	48.4	09/09	8.9770	347	2.8	54.4	321.8
T4173	A	46.277	238.280	59.3	19.3	130/14	71.8	4.7	09/09	8.9781	364	2.7	79.3	252.3
T4279[d,n]	A	46.521	238.890	-60.2	293.5	0/0	-60.2	293.5	09/09	8.9203	100	5.2	15.7	284.8
T4288	A	46.676	238.952	73.8	342.6	0/0	73.8	342.6	06/08	5.9618	131	5.9	73.4	207.4
T4296[d]	A	46.206	237.965	-30.9	110.9	25/05	-35.9	110.7	06/09	5.8984	49	9.6	28.4	147.7
T4305[o]	AT	46.204	237.945	45.3	11.6	25/05	46.2	16.7	09/09	8.9422	138	4.4	67.1	17.0
T4314	AT	46.197	237.950	54.3	2.9	25/05	55.9	9.7	09/09	8.9587	194	3.7	77.9	17.9
T4323[p]	SL	46.354	237.938	-77.1	177.6	355/19	-66.7	230.7	09/09	8.9835	484	2.3	56.5	304.0
T4332[d,p]	SL	46.343	237.884	68.2	82.9	355/19	49.2	83.8	09/09	8.5284	17	12.9	25.3	309.9
T4341[p]	SL	46.366	237.947	66.8	350.3	355/19	61.6	29.4	08/09	7.8367	43	8.6	68.9	327.1
Average				62.5	4.4	(k_G=26)	60.5	16.5	19/28	18.2726	k_S=25	6.9	77.5	330.5
T4001[d]	AB	46.552	238.388	20.6	3.7	290/25	-3.5	4.7	09/09	6.5007	3	34.4	41.5	52.1
T4010[d]	AB	46.552	238.384	-10.4	325.7	290/25	-24.0	319.0	06/08	1.9517	1	-.-	20.4	101.5

Site is laboratory identification number; unit is rock type: A, andesite flow; AB, andesitic breccia; AT, ash flow tuff; B, basalt flow; BA, basaltic andesite flow; D, dacite flow; DT, dacitic welded tuff: R, rhyolite flow or dome; RD, rhyodacite flow; and SL, Spirit Lake pluton; λ_S and ϕ_S are north latitude and east longitude of paleomagnetic sampling site in degrees; I_t and D_t are in situ (I_c, D_c are corrected) inclination and declination of mean paleomagnetic directions in degrees; strike/dip indicates the attitude of the rock unit with dip angles 90° clockwise of strike direction; N is the number of samples averaged; N_0 is the number of samples collected; R is the vector sum of N unit vectors; k is the concentration parameter [*Fisher*, 1953] and is also given for the uncorrected (k_G) and corrected averages (k_S); α_{95} is radius of 95% confidence in degrees; and λ_P and ϕ_P are north latitude and east longitude of corrected virtual geomagnetic pole in degrees. For directions with Bingham statistics [*Onstott*, 1980], R is not calculated, [k_1, k_2] are the two Bingham concentration parameters, and [α_1, α_2] are the two Bingham 95% confidence limits about the mean direction.

[a] Age of 30.1 ± 2.2 Ma (K-Ar, whole rock, MK98415 [*Phillips et al.*, 1986]).

[b] Age of 35.0 ± 2.0, 28.5 ± 1.1 Ma (K-Ar, plag), 33.5 ± 0.4 Ma (^{40}Ar-^{39}Ar, plag, S80-A3-E32 [*Evarts et al.*, 1994]).

[c] Age of 32.9 ± 2.6 Ma (K-Ar, whole rock, BP0814856 [*Phillips et al.*, 1986]).

[d] Omitted from further consideration due to random magnetizations [*Watson*, 1956].

[e] Just above flow with age of 25.5 ± 0.4 Ma (K-Ar, whole rock, MK85627 [*Phillips et al.*, 1986]).

[f] Age > 20 Ma (D. A. Swanson, unpublished data, 1994).

[g] Age 24.8 ± 0.3 Ma (U-Pb, zircon, JV-67); 26.5 ± 2.1 Ma (Fission track, zircon, JV-67 [*Vance et al.*, 1987]).

[h] Age of 24.4 ± 1.2 Ma (K-Ar, whole rock, BP0516851 [*Phillips et al.*, 1986]), 25.2 ± 0.6 Ma (K-Ar, whole rock, BP0516851 [*Evarts et al.*, 1994]).

[i] Age of 23.3 ± 0.8 Ma (K-Ar, plag, S81-D1-E19 [*Evarts et al.*, 1987]).

[j] Age of 22.8 ± 0.2 Ma (^{40}Ar-^{39}Ar, plag, 83FB-V06 [*Evarts et al.*, 1994]).

[k] Age of 15.7 ± 0.2 Ma (K-Ar, whole rock, MK85635 [*Phillips et al.*, 1986]).

[l] Age of 22.1 ± 1.3 Ma (K-Ar, whole rock, MK8578 [*Phillips et al.*, 1986]).

[m] Ages of 19.4 ± 1.0 Ma (K-Ar, plag, FRL#4817) and 20.0 ±1.0 Ma (K-Ar, plag, FEH-77-2 [*Hammond*, 1980]).

[n] Age < 23.3 ± 2.0 Ma (D. A. Swanson, unpublished data, 1994).

[o] Age of 24.9 ± 1.0 Ma (K-Ar, plag, 84SH-V82 [*Evarts et al.*, 1987]).

[p] Spirit Lake pluton, average age of 21 ± 1 Ma [*Evarts et al.*, 1987].

both normal- and reversed-polarity directions as expected in rocks of Tertiary age. Equal numbers of normal- and reversed-polarity sites also should be expected within each group, although rocks of normal polarity apparently have been preferentially sampled in groups 1 and 2. The presence of both polarities in these rocks indicates that they have not been regionally and pervasively overprinted and that they likely retain their original thermoremanent magnetizations.

Site-mean directions for characteristic magnetizations of the three groups (Figure 5) are dispersed, especially those of group 3. A certain amount of dispersion is expected due to PSV, and most directions should be within 30° to 40° of the averaged dipole direction. Those directions more widely dispersed could result from excursions or reversals of the geomagnetic field. Other possible sources of dispersion are inaccurate structural corrections, overlapping normal- and reversed-polarity components, and local vertical axis rotations of varying amounts. These other sources of dispersion, however, should equally affect groups 1 and 2, so it seems likely that behavior of the geomagnetic field is the cause of the high dispersion of group 3.

The reversal timescale for the middle Tertiary [*Harland et al.*, 1982] shows that the frequency of reversals increased between 36 and 16 Ma and that the frequency during deposition of group 3 rocks was twice that for group 1 rocks. Thus we suggest that the more widely dispersed site-mean directions for group 3 sites are due to reversals or excursions of the geomagnetic field during this time interval. These highly dispersed site-mean directions, with directions greater than 30° from the group's average direction (using all sites), have been omitted from the final group-mean calculations (Table 1). Only single sites have been omitted from groups 1 and 2, with totals of 21 and 26 sites respectively, but 9 sites out of 28 have been omitted from group 3. The corrected mean directions for group 3 be-

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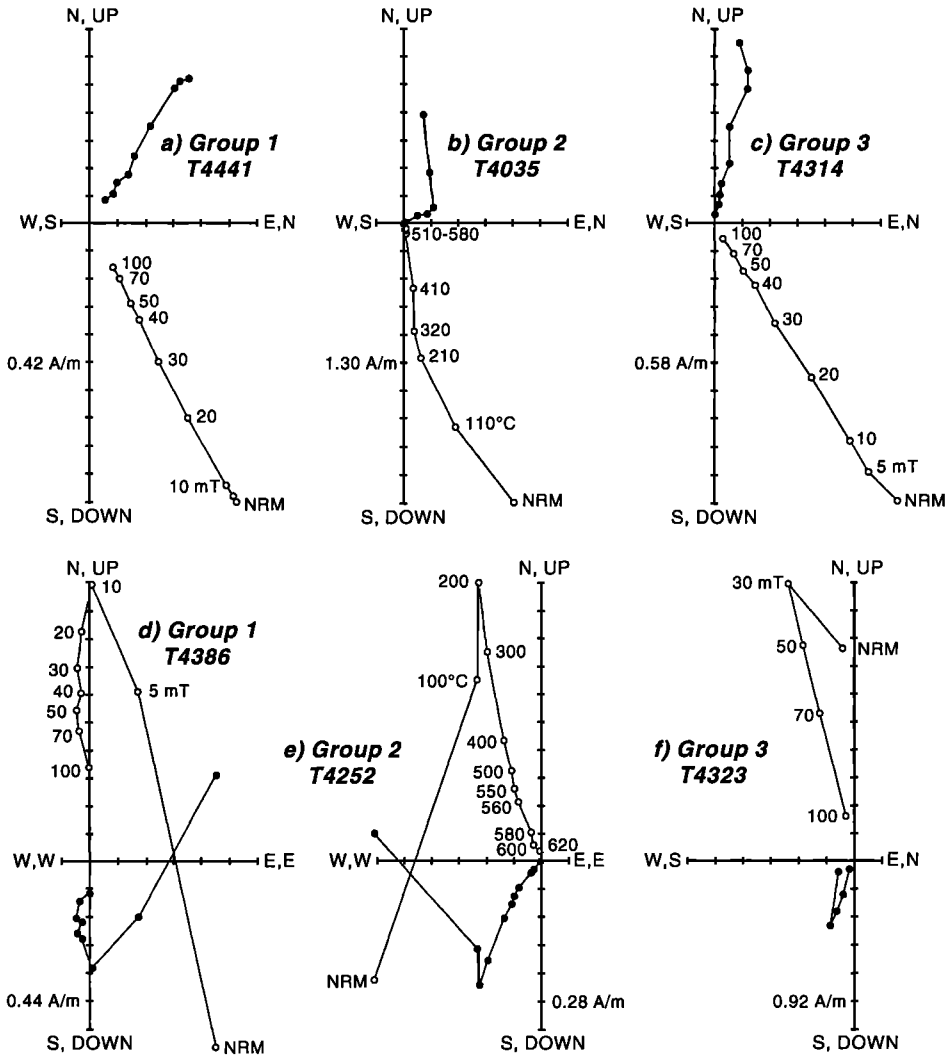


Figure 3. Orthogonal projections (uncorrected for tilt) of alternating field (AF) and thermal demagnetization vector endpoints for Cascade arc samples from groups 1, 2, and 3. (a), (b), and (c) Samples with normal-polarity characteristic magnetizations, and (d), (e), and (f) samples with reversed-polarity characteristic magnetizations. Figures 3b, 3d, 3e, and 3f show normal-polarity overprint magnetizations. Solid circles indicate projections onto the horizontal plane, and open circles indicate projections onto the vertical plane.

fore (I=59.4°, D=7.7°, N=28, k=8) and after (I=60.5°, D=16.5°, N=19, k=25) this selection are similar, however, indicating that the selection process has not strongly biased the mean direction.

Site-mean directions within each group pass reversal tests at the 95% confidence level. The angle between the normal- and reversed-polarity (inverted through the origin) mean directions (γ_c) for groups 1, 2, and 3 are 11.1°, 5.2°, and 14.2°, respectively, indicating positive reversal tests in the C, C, and B categories according to the classification of *McFadden and McElhinny* [1990]. In addition, the concentration parameter (k value, Table 1) for groups 1 and 2 have maxima at ~90-100% application of the structural corrections (Figure 6), indicating

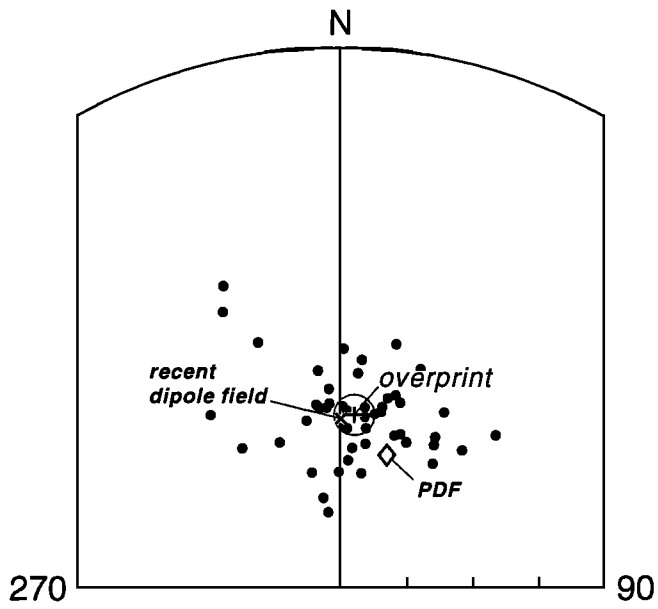


Figure 4. Equal-area plot of in situ normal-polarity overprint directions and their average (plus) with its 95% confidence limit. The present-day field (PDF) and recent dipole field (cross) directions are also shown.

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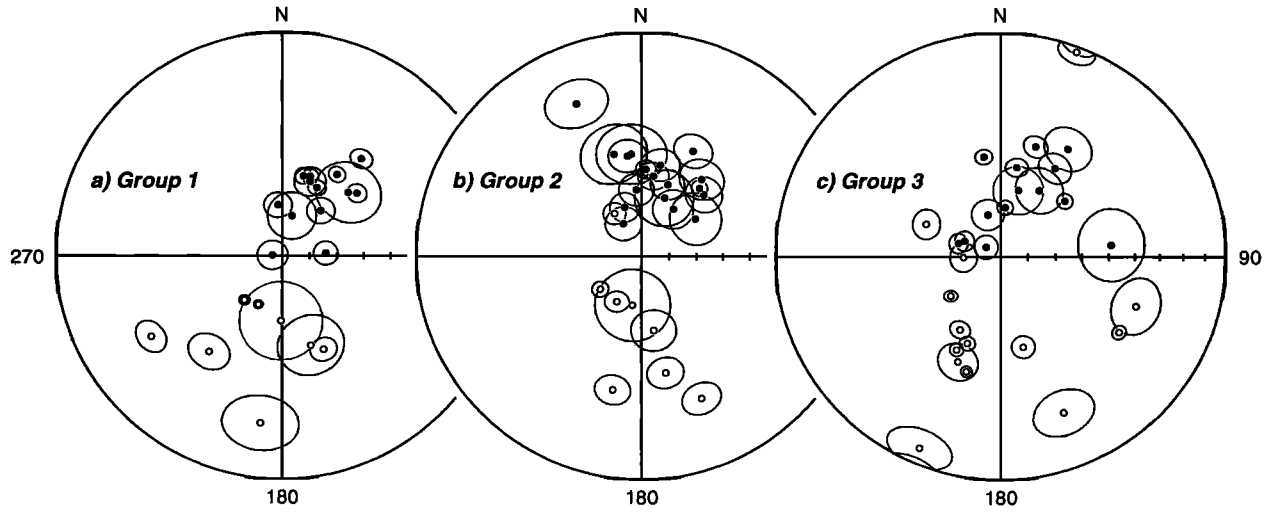


Figure 5. Equal-area plots of corrected site mean directions with their 95% confidence limits for the group 1, 2, and 3 samples. Solid (open) symbols indicate lower (upper) hemisphere points.

that these remanent magnetizations were likely to have been acquired prior to structural deformation. The concentration parameter for group 3, however, reaches a maximum at ~45% unfolding. The bedding corrections for group 3, including the youngest rocks, are relatively small and the variation in k values is insignificant within the error limits. Most likely the maximum k value at 45% unfolding is the result of inaccurate structural corrections in the extrusive rocks.

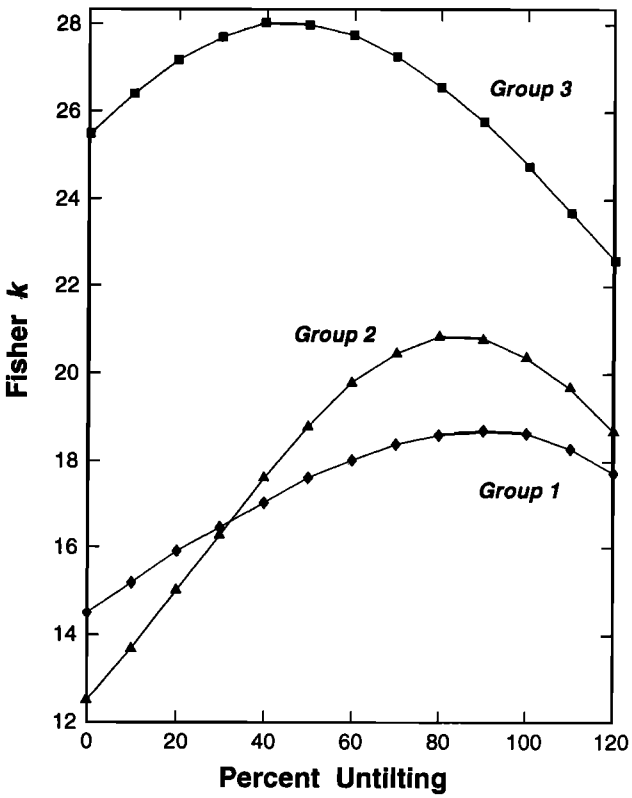


Figure 6. Plot of percent untilting versus the *Fisher* [1953] concentration parameter k for each of the three groups [*Watson and Enkin*, 1993]. Groups 1 and 2 show maxima at ~90-100% untilting, but group 3, including the youngest rocks, has a maximum at ~45% untilting (see text).

4.2. Regional Deformation

Clockwise rotations in the Pacific Northwest may be the result of two separate processes: dextral shear along the margin and the effects of Basin and Range extension [*Wells*, 1989; *England and Wells*, 1991]. Dextral shear decreases eastward away from the plate boundary, whereas the effects of Basin and Range extension decrease to the north from Oregon into Washington. The Miocene Snoqualmie batholith in northern Washington shows no vertical axis rotation [*Beske et al.*, 1973], and the position and nature of the boundaries between rotated and unrotated domains are unknown. Our data, however, extend the boundaries of vertical axis rotations in Miocene rocks northeastward to the axis of the arc in central Washington.

The rates of rotation in southern Washington (minimum rates calculated by dividing the rotation angle by the rock's or unit's mean age) are also similar to rates of rotation for volcanic rocks of the Western Cascades in Oregon (Table 2). *Magill and Cox* [1980] report clockwise rotations of 30° for two sites in the Oligocene (30-20 Ma) Western Cascades Volcanics which also corresponds to a rate of ~1°/m.y. Southern Washington is beyond the influence of Basin and Range extension and the rotations there are most likely the result of dextral shear. In Oregon, however, the relatively aseismic Oregon block, which includes the Western Cascades, is moving clockwise relative to stable North America at an overall rate of ~1.5°/m.y. and is creating an extensional volcanic arc at its trailing edge [*Wells et al.*, 1998]. This motion is the sum of both dextral shear and Basin and Range extension, although there is little evidence of dextral shear within the Oregon block.

Group-mean poles for the southern Washington transect (Table 2) are shown in Figure 7. Also shown are reference poles for the Oligocene to early Miocene [*Diehl et al.*, 1983] and Miocene [*Hagstrum et al.*, 1987], and the mean in situ pole for the intrusive suite of Kidd Creek [*Hagstrum et al.*, 1998]. The group-mean and Kidd Creek poles are all deflected clockwise from the middle Tertiary reference poles with respect to the sampling locality in southern Washington, indicating that these rocks have been rotated about vertical or near vertical axes. Values of clockwise rotation (Δr) and northward latitudinal displacement (Δd) are given in Table 2. The clockwise ro-

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Table 2. Observed Directions, Reference Poles, and Displacement Parameters for Volcanic and Intrusive Rocks of the Cascade Arc, Southern Washington

Group	I	D	α_{95}	λ_{RP}	ϕ_{RP}	A_{95}	Δd	Δr	Rate
GVE	57.0	19.9	4.4	83.2	148.0	4.1	7±4	30±8	0.8
OV	64.7	25.2	4.1	83.2	148.0	4.1	-1±4	35±9	1.0
1	59.6	24.6	7.8	83.2	148.0	4.1	5±7	34±13	1.0
2	59.2	7.2	6.6	83.2	148.0	4.1	5±6	17±11	0.6
3	60.5	16.5	6.9	87.4	129.7	3.0	3±6	20±12	1.0
KC	66.6	18.1	3.9	87.4	129.7	3.0	-3±4	22±8	1.7
SHZ	60.4	22.7	2.9	62.2	13.2	3.4	2±3	10±3	-.-

GVE, Goble Volcanics [*Beck and Burr*, 1979]; OV, Ohanapecosh Volcanics [*Bates et al.*, 1981]; 1, 2, and 3, groups of this study; KC, intrusive suite of Kidd Creek [*Hagstrum et al.*, 1998]. I, D, and α_{95} are inclination, declination and 95% confidence limit of mean direction in degrees for unit or group. λ_{RP}, ϕ_{RP}, and A_{95} are latitude, longitude, and 95% confidence limit of reference pole in degrees (see text). Δd is northward latitudinal displacement in degrees with 95% confidence limits, and Δr is clockwise rotation in degrees with 95% confidence limits, both errors calculated using the method of *Demarest* [1983]. Rate is rotational rate in degrees per million years. SHZ is the St. Helens seismic zone, and the mean directions and confidence limits for site groups west and east of the zone are listed from left to right.

tations are significant at the 95% confidence level, and the latitudinal displacements are not indicating that the vertical-axis rotations occurred in place.

The group 1 pole (Figure 1) shows the greatest clockwise rotation (34° ± 13°) which is consistent with regional models having older rocks associated with greater rotations [e.g., *Wells*, 1990]. The group 2, group 3, and Kidd Creek poles have lesser and roughly equivalent values of clockwise rotation between about 17° and 22° (Table 2). All three of these group-mean poles, however, are indistinguishable at the 95% confidence level [*McFadden and Lowes*, 1981]. Thus there are several possible interpretations of the data for the southern Washington arc: (1) the total regional rotation occurred after emplacement of the intrusive suite of Kidd Creek (13 Ma), (2) ~10° of clockwise rotation occurred between deposition of group 1 and group 2 rocks and another 20° of rotation occurred

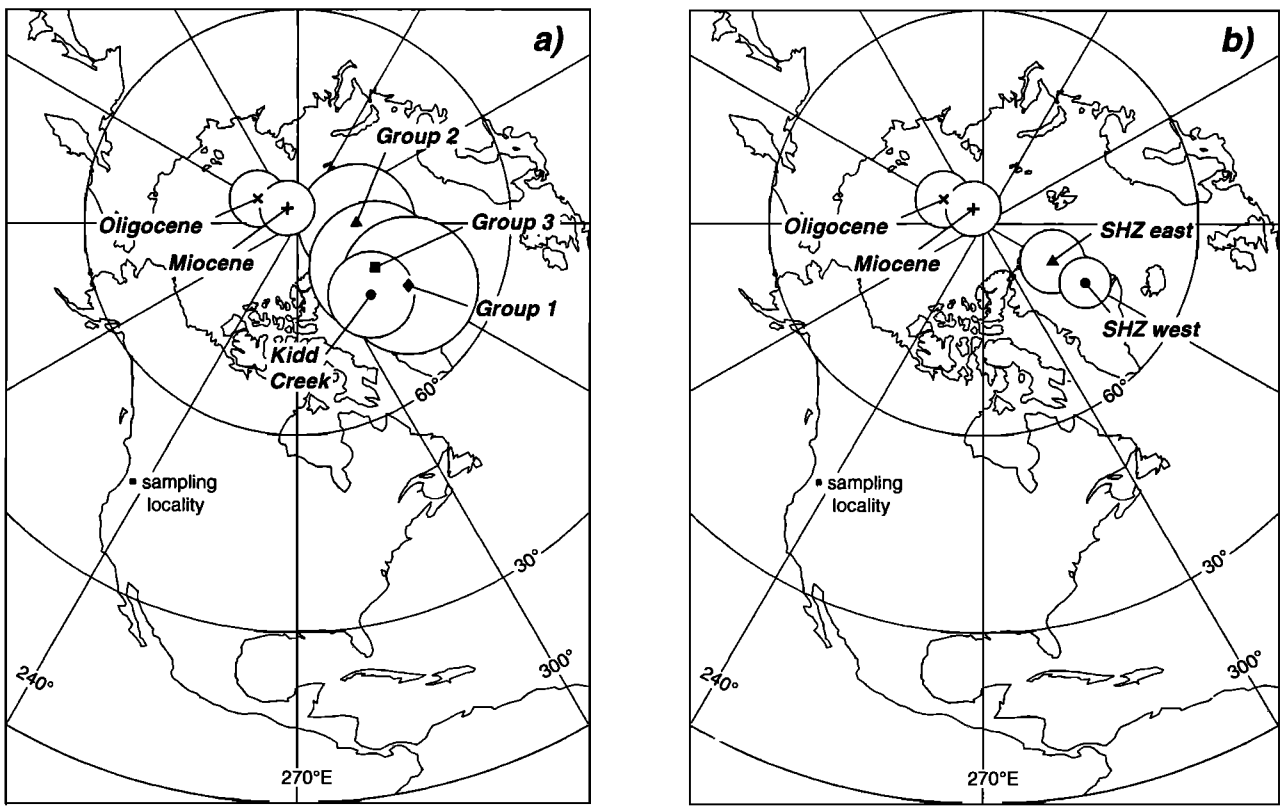


Figure 7. (a) Lambert equal-area map of group-mean poles and reference poles for Cascade Range samples in southern Washington. (b) The mean poles for samples east and west of the St. Helens seizmic zone.

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after emplacement of the intrusive suite of Kidd Creek, or (3) the difference in rotation between groups 1 and 2 across the St. Helens seismic zone (SHZ), and similar rotations for groups 2 and 3 east of the SHZ, results from differential rotation across this crustal structure (Figure 2).

In this study the age break between groups 1 and 2 at ~30 Ma roughly coincides with sampling sites west and east, respectively, of the SHZ, so it is difficult to determine whether this break results from a geographic or age effect. However, groups 2 and 3 and the intrusive suite of Kidd Creek show similar vertical axis rotations despite their different ages, which support a model of differential rotation across the SHZ. Furthermore, virtually all of the sites sampled by *Beck and Burr* [1979] and *Bates et al.* [1981] are west of the SHZ, or its southward projection, so the terrane west of the seismic zone apparently has been rotated more than the terrane to the east, consistent with the regional trend of larger rotations toward the coast [*Beck*, 1980; *England and Wells*, 1991]. Combining the site-mean directions for group 1 with those from the studies by *Beck and Burr* [1979] and *Bates et al.* [1981] west of the SHZ (I=60.4°, D=22.7°, N=90, α_{95}=2.9°) and comparing them with a combination of site-mean directions for groups 2 and 3 and the intrusive suite of Kidd Creek [*Hagstrum et al.*, 1998] (I=62.2°, D=13.2°, N=68, α_{95}=3.4°) indicates a differential rotation across the SHZ of 10° ± 3° (Table 2).

The seismicity, aeromagnetic anomalies, magnetotelluric anomaly, and seismic reflection studies all indicate that the SHZ is an important boundary between crustal blocks, and the paleomagnetic data indicate that it is a rotational domain boundary as well. Much more work, however, is needed to unravel the complex details of rotational deformation in the Pacific Northwest.

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